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CIDM 6325 Assignment 1

Project Pitch: Real Time Fraud Dashboard (RTFD)

Problem

E-commerce platforms continue to face rising fraud losses, particularly in the areas of account takeover and payment fraud. Global fraud losses exceeded $50 billion in 2023, underscoring the scale of the challenge (Juniper Research, 2023). Many existing fraud defenses rely on static rule sets that struggle to adapt to rapidly evolving attack techniques.

Company X currently lacks a real-time fraud monitoring capability that integrates user behavior analytics with transaction data. As a result, high-risk orders are identified too late, false positives create unnecessary customer friction, and analysts lack a centralized view of fraud activity.

RTFD seeks to close this gap by delivering a lightweight but functional prototype that applies baseline fraud scoring to recent transactions and provides analysts with a dashboard for timely triage.

Stakeholders

- **Fraud Operations Team –** Requires faster triage of suspicious orders with clear prioritization.
- **Customer Success and Support Teams –** Need fewer false positives to reduce friction and maintain customer trust.
- **Merchants and Business Partners –** Rely on trust, low chargeback rates, and streamlined order processing.
- **Executive Leadership –** Seeks reduced fraud costs, improved compliance alignment, and stronger risk visibility.

Scope (2-week sprint)

In-scope

- Develop a prototype dashboard ingesting a sample dataset (~10,000 recent transactions).
- Implement a baseline fraud scoring model using simple but interpretable features (IP geolocation mismatch, device fingerprint irregularities, velocity checks).
- Provide an interface highlighting the top 50 high-risk transactions flagged in near real time.

- Deploy in a lightweight, test-ready environment.

Out-of-scope

- Full production-ready machine learning pipelines with automated retraining.
- Integration with payment processors at production scale.
- Formal legal or regulatory certification.

Minimal Viable Artifact (MVA)

The prototype delivers:

1. A functional dashboard displaying fraud scores for live test transactions.
2. A baseline scoring engine capable of identifying high-risk patterns.
3. An analyst workflow that supports triage of the top 50 flagged transactions.

This artifact is intentionally modest in scope but addresses key operational blind spots, demonstrates early value, and establishes a foundation for future iterations.

Success Metrics

- **Detection coverage**: At least 80% of known fraudulent test cases are flagged.
- **False positives**: Legitimate orders incorrectly flagged remain below 20% on validation data.
- **Usability**: At least three fraud analysts complete test workflows successfully and report positive feedback.
- **Latency**: Fraud scores generated in under one second per transaction.

Iterative Design Approach

The project follows a short, iterative cycle of design, implementation, testing, and stakeholder feedback. The initial sprint emphasizes transparency and usability—providing a working dashboard with baseline scoring. Subsequent iterations may extend into more advanced analytics, automated retraining, and deeper integration, depending on results and stakeholder priorities.

System Sketch

```
+------------------+
|   User Checkout  |
+------------------+
         |
         v
+------------------+
| Front-End API    |
| (captures order) |
+------------------+
         |
         v
+------------------+
| Transaction DB   |
| (orders, users)  |
+------------------+
         |
         v
+----------------------------+
| Fraud Scoring Engine       |
| - Rules (IP mismatch, etc.)|
| - Baseline ML model        |
+----------------------------+
         |
+-------------------------------------------+
|                                           |
v                                           v
+-------------------------------+   +--------------------------------+
| Fraud Dashboard UI  |             |  Evidence Drawer   |
| - Top 50 flagged    |             | - Signals/thresholds  |
| - Filters & search  |             | - Assumptions logged  |
| - Analyst actions   |             | - Transparency data   |
+-------------------------------+   +-----------------------------+
         |
         v
+----------------------+
| Analyst Decision Log |
| (approve/hold/decline) |
+----------------------+
```

Evidence Base

- Juniper Research (2023). *Online Payment Fraud: Emerging Threats & Market Forecasts 2023–2027.*
- Association of Certified Fraud Examiners (2022). *Report to the Nations: Global Study on Occupational Fraud and Abuse.*
- National Institute of Standards and Technology (2020). *SP 800-53 Rev. 5: Security and Privacy Controls for Information Systems and Organizations*.

Risk Register

Risk	Likelihood	Impact	Mitigation
Data quality issues (missing or inconsistent fields)	Medium	High	Validate schema before ingestion; supplement with synthetic/mock data.
Model underperformance (baseline too simple for real-world fraud patterns)	Medium	Medium	Benchmark against rules-only baseline; frame as prototype; plan for iterative model improvements.
Stakeholder resistance (analysts hesitant to adopt new tool)	High	Medium	Involve fraud analysts early; gather feedback during sprint reviews; iterate based on usability input.

Closing

This two-week sprint will deliver a fraud monitoring prototype that directly addresses critical gaps in visibility and responsiveness. By centralizing suspicious activity into a single dashboard, the system reduces operational blind spots and equips fraud analysts with the tools to act quickly on high-risk orders. The prototype also improves triage efficiency by ranking transactions by risk level, giving teams the ability to prioritize workload and focus on the most pressing threats.

While intentionally limited in scope, the Real-Time Fraud Dashboard represents a meaningful first step toward a more mature fraud prevention ecosystem. It not only demonstrates immediate value to stakeholders by surfacing actionable insights but also establishes a framework that can support future growth. Over time, the prototype can be extended with advanced machine learning models, automated retraining pipelines, and tighter integration with payment processors and customer support systems. In this way, the MVP lays a practical foundation for enterprise-level scaling and continuous improvement, ensuring the organization can adapt to evolving fraud tactics while maintaining customer trust and operational resilience.